CoRoT-2b

R-1623

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_210708` · created 2026-08-02T04:50:24+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459403.7798 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.224 +/- 0.065
Transit depth (Rp/Rs)^2	5.02 +/- 2.93 [%]
Orbital Inclination (inc)	89.42 +/- 2.07
Airmass coefficient 1 (a1)	0.81 +/- 0.42
Airmass coefficient 2 (a2)	-0.06 +/- 0.68
Scatter in the residuals of the lightcurve fit is	53.43 %
Best Comparison Star	#3 - [302, 370]
Optimal Aperture	2.64
Optimal Annulus	3.0
Transit Duration (day)	0.0999 +/- 0.0062

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.224 ± 0.065 vs 0.1667 (deviation 0.88σ)
Duration fit vs published	0.0999 d vs 0.095 d (deviation 0.79σ)
Mid-time O–C	0.3 min
Depth SNR	3.4
Residual scatter	53.43 %

Light curve

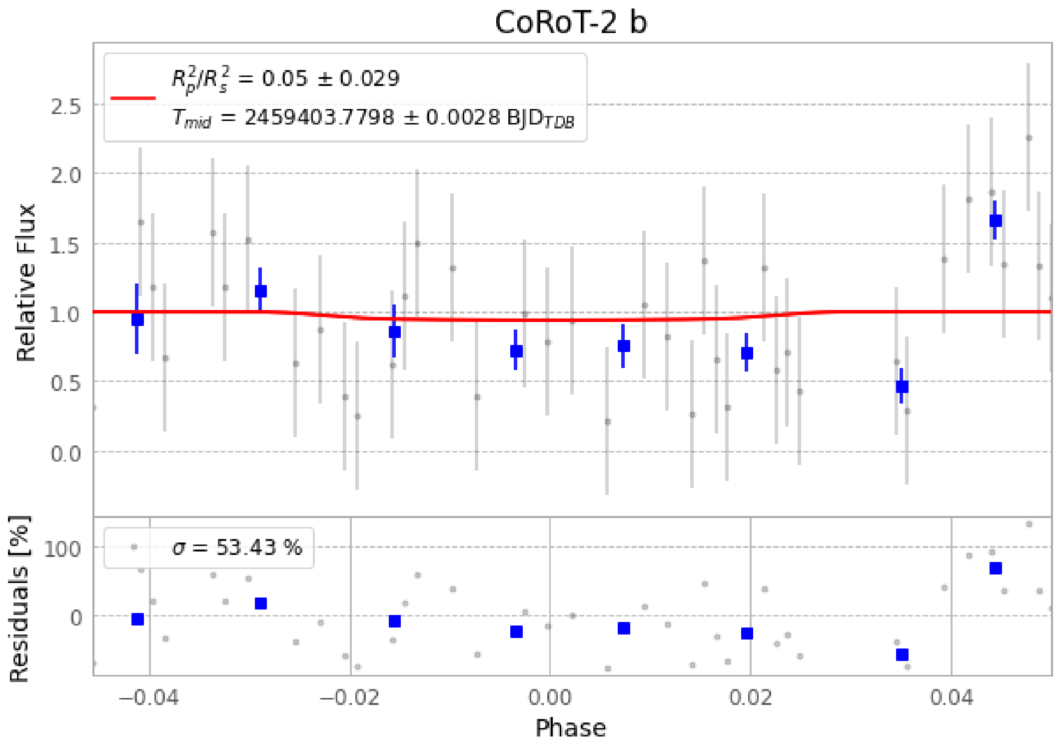

Figure 1 — FinalLightCurve_CoRoT-2 b_2021-07-07.png (SHA-256 2a30f4bd3bd06bc1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [256, 394], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 b229e9d26251d71c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

85 FITS frames (56 MB), `CoRoT-2210708043012.FITS` → `CoRoT-2210708084212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-210708.log (08649f11e69d…), FinalLightCurve_CoRoT-2 b_2021-07-07.png (2a30f4bd3bd0…), FinalLightCurve_CoRoT-2 b_2021-07-07.csv (c1af7110297f…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)